

PPT content kit

Problem statement SIH26099 · CPCL, Ministry of Petroleum & Natural Gas · Team Overwatch

What to put on each of the six slides, what to say, which picture goes where, and the source of every fact. Every number comes from the locked run (git tag `ppt-numbers-2026-09-25`); every outside fact was checked against the source linked in the References section.

HOW TO USE IT

1. Open the official template `SIH2026-IDEA-Presentation-Format.pptx`. **Keep its headings exactly** — the template says the idea pointers must not be changed.
2. For each slide, copy the text in the blue **“Put on the slide”** box. It is already in points, not paragraphs.
3. Insert the picture named under **Visual**. All pictures are in the `images` folder next to this PDF.
4. The orange **“Say”** notes are for the presenter, not the slide.
5. Delete the template's “Important instructions” slide, export as **PDF**, six slides maximum, and upload.

THE TEMPLATE'S OWN RULES

- Six slides maximum, including the title slide. Save as PDF — no PPT or Word upload.
- Points, diagrams, infographics and pictures — avoid paragraphs. Keep it precise and easy to understand.
- Use only the provided template; don't change the idea-detail pointers.

WHAT THE JUDGES SCORE, AND WHERE WE ANSWER IT

CRITERION (SIH EVALUATION)	WHERE THE DECK ANSWERS IT
Novelty · innovation	Slide 2 “Innovation and uniqueness” — specifications not spelling, standards built in, every refusal explained
Problem understanding	Slide 2 — real NTPC / Oil India tender lines, the SAP 40-character field, 272 CPSEs
Complexity · technical depth	Slide 3 — the architecture diagram, where AI is used and where rules decide
Feasibility · practicability	Slide 4 — a working prototype with measured numbers; risks with a strategy for each
Scale of impact · sustainability	Slide 5 — savings, safety, the creation gate that stops new duplicates
Clarity · completeness	Every slide in points; slide 6 references every source

Criteria as published for SIH evaluation: novelty, complexity, clarity, feasibility, practicability, sustainability, scale of impact, user experience, potential for future work.

THE LOCKED NUMBERS AT A GLANCE

0 wrong

of 860 merges made with no person
(15,000 records)

< 0.35%

wrong-merge rate, 95% confidence
(Clopper–Pearson)

0 of 104,994

planted near-miss traps merged

82%

of duplicates found, with a reviewer

6–8%

right: text matchers at their best
setting

0 wrong

on 252 real NTPC & Oil India lines (4
merged, 83% of duplicates found
with a reviewer)

PUT ON THE SLIDE

Problem Statement ID	SIH26099
Problem Statement Title	AI-Driven Standardization and Harmonization of Material Codes Across CPSEs
Theme	Smart Automation
PS Category	Software
Team ID	from the SIH portal
Team Name (registered on portal)	Overwatch

OPTIONAL, IF THE TEMPLATE LEAVES ROOM

Idea name under the title: **One Nation, One Material Code** — *matching refinery materials on specifications, not spelling.*

Say: "One idea runs through everything we built: two material records are the same item when their specifications agree, not when their words look alike."

Visual: none needed — the template's title design is enough. If a picture is wanted, use `photo-refinery.jpg` (an Indian PSU refinery, credit on slide 6).

IDEA TITLE**One Nation, One Material Code — match materials on specifications, not spelling****DETAILED EXPLANATION OF THE PROPOSED SOLUTION**

- **Read** every CPSE material description into typed specifications — size, grade, pressure class, standard — keeping the exact words each value came from.
- **Match** on specifications; issue **one National Material Code**; keep every company's own code, mapped. Nothing is deleted.
- **Refuse safely**: one mismatch on a veto field (grade, pressure class, size) and the merge is refused — with the reason.

HOW IT ADDRESSES THE PROBLEM

- Finds the same item written differently, inside a company and across CPSEs: *"Line Pipe 101.60 mm ... 12.66 PPF"* and *"Line Pipe, 4" ... 12.66 PPF"* are one pipe — the weight proves it.
- Stops new duplicates: a **creation gate** checks every new request before a code is issued.
- Gives each company a **mapping file for its SAP team**, and procurement a view of what CPSEs buy in common.

INNOVATION AND UNIQUENESS

- **Specifications, not spelling.** Text matchers are 6–8% right at their best; ours made 0 wrong merges.
- **Engineering standards built in:** ISO 15 bearings, ASME B36.10 pipe, IS 1367 = ISO 898-1, withdrawn ASTM grades mapped to F3125.
- **Every refusal explained:** "if the grade matched, this pair would score 0.956."
- **Catches a line that contradicts itself** — a "6205" bearing written with a 6206's size.
- **Federated and governed:** prices and vendors never leave a company raw; any company can dispute a link; every decision is sealed in an audit log.

VISUAL

`inversion.png` (the centre of the slide) — optionally `photo-ball-bearing.jpg` and `photo-gate-valves.jpg` as small corner pictures.

Text similarity gets both pairs backwards

So we match on specifications, not spelling.

SAME BEARING

few shared words, same part

SKF 6205-2RS DEEP GROOVE BALL BEARING

FAG 6205-2RSR BEARING, 25X52X15MM, SEALED

Text similarity

0.833

a text matcher is unsure

Specifications, field by field

✓ 6205 = 6205
✓ 25 × 52 × 15 mm from ISO 15
✓ 2RS = 2RSR (sealed)
✓ brand ignored

MATCH 0.971

DIFFERENT STEEL

one character apart, never interchangeable

GASKET SPIRAL WOUND SS316 4IN 150#

GASKET SPIRAL WOUND SS304 4IN 150#

Text similarity

0.989

a text matcher says: same

Specifications, field by field

✗ SS316 ≠ SS304 (veto)
✓ 4 in = 4 in
✓ 150# = 150#
↳ if the grade matched: 0.956

REFUSED 0

The pair that must never merge scores HIGHER on text (0.989) than the pair that should (0.833). No threshold fixes an inverted order.

inversion.png — the core idea in one picture. Real output of our system on the demo pair.

Say — the problem (20 seconds). “India has 272 operating CPSEs. Each keeps its own material master. The same gate valve gets a different code and a different description in every company — so nobody can see that five companies buy the same item, and duplicates pile up in stores.”

Why descriptions are so messy — say it this way. “Several reasons: SAP's standard description field is still 40 characters, data came over from older systems, and above all there was never an enforced cataloguing standard — so every storekeeper invented their own shorthand.” **Never** blame SAP alone; a CPCL judge will know the real cause is the missing cataloguing policy.

The inversion (the line to remember). “Two records for one bearing share almost no words. Two records for different steel share almost all of them. Text matching scores the wrong pair higher. No threshold can fix an inverted order — so we changed what is measured.”

Why grade matters. “SS316 carries 2–3% molybdenum; SS304 has none. In chloride service the wrong one pits and leaks.”

Sources: 272 operating CPSEs — DPE Public Enterprises Survey 2023-24. MAKTX = 40 characters — SAP KBA 1630702. SS316 molybdenum — ASTM A240 / supplier data. Real pipe lines — Oil India tenders in the Hugging Face dataset (CC-BY-4.0). See References.



photo-ball-bearing.jpg — R. Henrik Nilsson, CC BY 4.0



photo-gate-valves.jpg — Bitjungle, CC BY-SA 4.0

TECHNOLOGIES TO BE USED

- **Python pipeline** — normaliser, rule reader, standards tables (ISO 15, ASME B36.10), scorer, clustering
- **LLM** (Gemini / Claude) — reads a description **once per record**, never per pair; rules run first, offline
- **Local embeddings** — sentence-transformers, 384-dimension vectors; no internet needed
- **Scale:** blocking with identity keys (skips 99.8% of pairs); PostgreSQL + pgvector at national scale
- **Portal:** React + Tailwind, government style, role-based; runs offline; FastAPI service in the next phase
- **Trust:** 292 automated tests · SHA-256 sealed audit log · UNSPSC classification

METHODOLOGY AND PROCESS FOR IMPLEMENTATION

- Flow chart: `architecture.png` (full width) — eight stages, three lanes: read · match · registry
- **The AI reads; the rules decide.** AI in two places: an LLM reads each record once, and embeddings find candidates. Every merge is decided by rules and can be explained. A statistical bound measures the error (< 0.35% at 95%); it does not limit it.

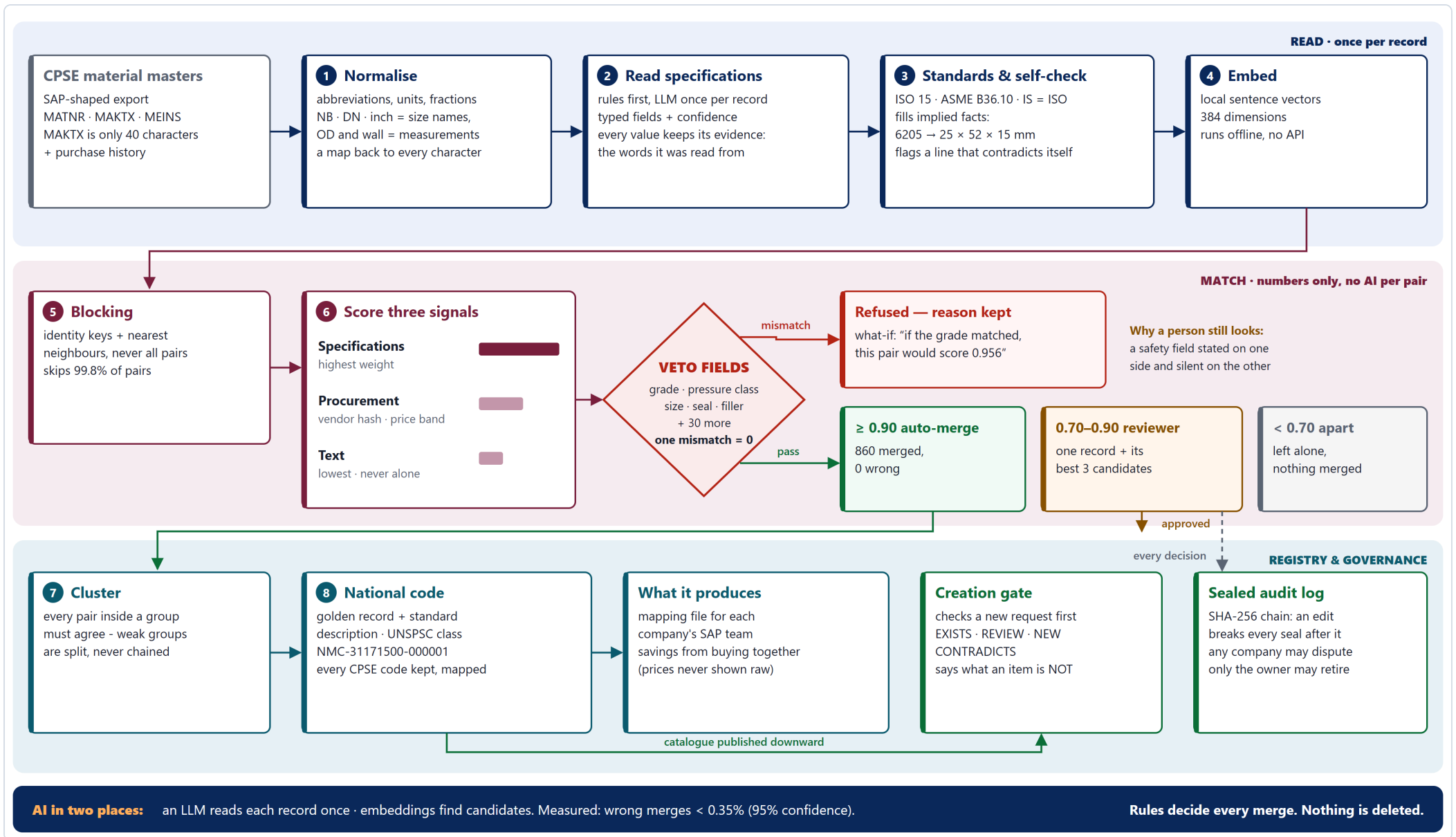
WORKING PROTOTYPE (ONE SCREENSHOT)

`screen-review-queue.png` OR `screen-real-text.png`

Say. “Three stages. First we read: every description becomes typed specifications, once, with the words each value came from. Then we match — with numbers only, no AI per pair, so it is cheap and every decision can be replayed. Safety fields veto a merge outright. Last, the registry: one national code, every company code kept, and a sealed log of who decided what.”

If asked “why not just use ChatGPT to match?” “An LLM per pair is millions of calls, costly, and not reproducible. We call it once per record to read, and let rules decide — so a judge, an auditor or a CPSE can check every merge.”

Be accurate about the stack: the pipeline, portal, tests and audit log are built and running; FastAPI, PostgreSQL/pgvector and Docker are the decided production stack, not yet built. Say “next phase” for those.



3 PROTOTYPE SCREENSHOTS — pick one for slide 3

record 1 of 36 5 CANDIDATES 2 MORE AFTER THESE

WHICH OF THESE IS THIS ITEM?

IOCL MAT-01643 NO

MOTOR - INDUCTN - 3PH - 5 HP - 1440 RPM - FOOT MOUNED - IP55

1 GAIL G/001697 7 FACTS CONFIRMED SCORE 0.89 ACROSS COMPANIES

MOTOR, INDUCTN, 3PH, 5 HP, 1440 RPM, FOOT MOUNTED, IP55, 415 V, ABB

could not verify voltage v - not extracted from both records. Confirm voltage (v) and it scores 0.984.

2 GAIL G/001721 7 FACTS CONFIRMED SCORE 0.89 ACROSS COMPANIES

Motor, Induction, 3-Ph, 5 Hp, 1440 Rpm, Foot Mtd, Ip55, 415 V

could not verify voltage v - not extracted from both records. Confirm voltage (v) and it scores 0.972.

3 ONGC 5001646 7 FACTS CONFIRMED SCORE 0.89 ACROSS COMPANIES

Motor - Induction - 3 Phase - 5 Hp - 1440 Rpm - Foot Mounted - Ip55 - 415 V - Kirloskar - Mtr-5Hp-1440-B3

could not verify voltage v - not extracted from both records. Confirm voltage (v) and it scores 0.972.

screen-review-queue.png — the reviewer sees one record and its best three candidates, ranked by how many facts agree, with the reason a person is needed and what confirming it would score.

Oil India line 163 read as pipe

Line Pipe 101.60 mm Seamless, B/E, 19 PPF. Qty:40,000 Mtr

Weight (lb/ft) 19 Nominal size (in) 4 Wall (mm) 11.13 Construction SEAMLESS Ends BEVEL Outside diameter (mm) 114.3 · from the standard

Oil India line 329 read as pipe

Line Pipe, 4", Seamless, BE, 19 PPF

Weight (lb/ft) 19 Nominal size (in) 4 Construction SEAMLESS Ends BEVEL Outside diameter (mm) 114.3 · from the standard

Wall (mm) 11.13 · from the standard

SENT TO A REVIEWER 0.89 words in common 31% agreed on construction, nominal size (in), outside diameter (mm), wall (mm), ends, weight (lb/ft)

Why a person still looks: could not verify material grade - not extracted from both records.

screen-real-text.png — real Oil India lines: highlighted words are where each value was read; dashed tags were filled in from a standard. "101.60 mm" and "4"" are matched through the stated weight.

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Working prototype, measured** — runs end to end on a laptop, offline. Numbers strip: `numbers.png`
- **15,000 records** written the way NTPC, Oil India and SAP write: **860 merged automatically, 0 wrong** (below 0.35% at 95% confidence); **0 of 104,994** near-miss traps merged; 82% of duplicates found with a reviewer.
- **Beats text matching** on the same pairs (chart `text-vs-specs.png`): fuzzy, TF-IDF and embeddings are 6–8% right at their best; no text threshold reaches zero wrong.
- **Real tender text** — 252 held-back Oil India and NTPC lines: item type read right 98%, **0 wrong merges**, 83% of true duplicate pairs found with a reviewer.

POTENTIAL CHALLENGES AND RISKS → STRATEGIES FOR OVERCOMING THEM

CHALLENGE	STRATEGY
No CPSE will share prices or vendor lists	Federated: only price bands and vendor hashes cross the boundary; a benchmark is hidden below 3 companies. Phase 1 — duplicates inside one company — needs no sharing at all.
A wrong merge can put the wrong valve in a line	Veto fields; a safety field unstated on one side goes to a person; self-check; a measured error bound; people decide every uncertain pair.
Descriptions are cryptic and inconsistent	Normaliser + engineering standards + evidence for every value; tested on real tender text, not only on our own data.
Millions of records	Blocking skips 99.8% of pairs; the LLM runs once per record, never per pair.
Reviewers do the work, others get the benefit	Only genuinely uncertain pairs reach a person; one record with its top three candidates; keyboard only; the gate stops new mess.
Our generated data could flatter us	A real hold-out, labelled separately by two people, is reported beside it — never blended.

Say — the honest line (it wins trust). “Our 15,000-record numbers measure how well we undo noise we generated ourselves. That is why the real tender test sits beside them. Its weak point is real: only about one in five pairs we send a reviewer is a true duplicate, because a short line fits several items. So the reviewer sees one record with its best three candidates — the right one is in the top three about four times in five.”

Numbers: `data/output/evidence.json` at tag `ppt-numbers-2026-09-25` (see `deliverables/slide-numbers.md`). Error bound: Clopper & Pearson (1934). Real test labels are so far one person's (the builder's); two team members are labelling separately — say so if asked.

0 wrong

of 860 automatic merges

< 0.35%

wrong-merge rate, 95% confidence

0 of 104,994

planted near-miss traps merged

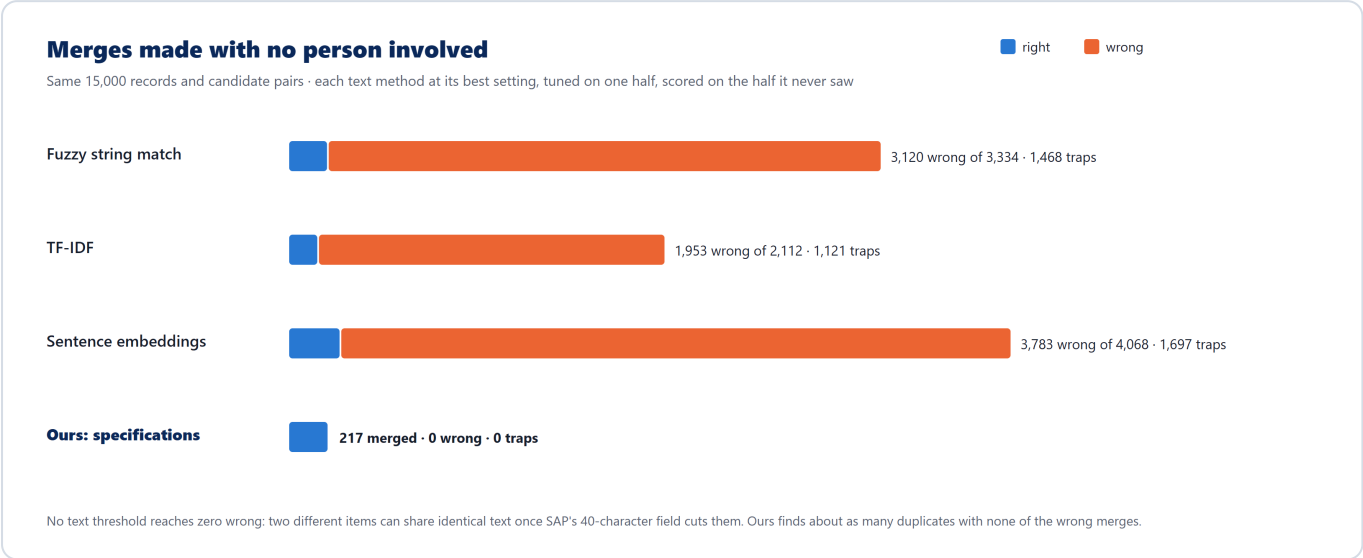
82%

of duplicates found, with a reviewer

0 wrong

on real NTPC & Oil India text (4 merged)

numbers.png — the headline strip for slide 4 (or type these numbers directly).



text-vs-specs.png — same 15,000 records and candidate pairs; each text method tuned on one half and scored on the half it never saw. Blue = right merges, orange = wrong.

METHOD (FAMILIES NEVER USED FOR TUNING)	MERGED	WRONG	TRAPS MERGED	RIGHT
Fuzzy string match	3,334	3,120	1,468	6%
TF-IDF	2,112	1,953	1,121	8%
Sentence embeddings	4,068	3,783	1,697	7%
Ours — specifications	217	0	0	all

POTENTIAL IMPACT ON THE TARGET AUDIENCE

- **Cataloguers / reviewers** — decide only the genuinely uncertain pairs, with the evidence on one screen.
- **Storekeepers** — a new request is checked before a code is issued; the gate says what an item is *not*.
- **Procurement officers** — see what several CPSEs buy in common: joint tenders, better prices.
- **CPSE SAP teams** — a ready mapping file: company code → national code.
- **MoPNG / national registry** — one national view, savings and a governance trail.

BENEFITS OF THE SOLUTION

- **Economic:** published cases report ~5–7% lower inventory cost after de-duplication, and ~10% potential MRO savings at an oil & gas company. On our demo data: ₹2.31 cr upper / ₹0.92 cr realistic saving on ₹22.6 cr of spend.
- **Safety:** 13,127 unsafe merges refused that were one field from happening — SS304 kept apart from SS316.
- **Operational:** duplicates stopped at creation; every merge explained, reversible, and sealed in an audit log.
- **National:** an Indian registry on BIS / ISO standards; company data stays with the company.
- **Environmental:** less duplicate stock means less dead inventory later written off and scrapped.

VISUAL

screen-evidence-panel.png (portal dashboard) or screen-gate-contradicts.png

Say. “CPSEs bought more than ₹91,000 crore through GeM in a single quarter. When one item carries a different code in every company, none of that spend can be compared. A national code makes it comparable — that is where joint buying comes from.”

Always attribute the percentages — “published case studies report”, never “we will save 5–7%”. The ₹ figures are computed on synthetic prices: they show the method, not a forecast.

Sources: CODA Technology Solutions case study (5–7% inventory cost); EY MRO master-data case study (10% potential MRO savings, 1M+ records); GeM Q1 FY25 CPSE procurement (PSU Watch / PTI, 11 July 2024). See References.

How we know it is safe

Measured on three datasets, side by side and never blended. Every number is read from the run that produced it.

15,000 RECORDS - REAL NOTATION

Generated from standards, written the way NTPC, Oil India and SAP write

0 wrong

of 860 merges made with no person involved

Wrong-merge rate, 95% confidence **< 0.35%**

Families held back from tuning **0 of 217**
3,336 records, two notations never seen

Near-miss traps merged **0 of 1,04,994**
SS304 vs SS316, 150# vs 300#, 6205 vs 6305 ...

Refused one field from a merge everything else agreed **13,128**

Duplicates found, with a reviewer **82%**

Right match in the reviewer's top 3 **78%**

data/runs/15k/output/metrics.json · 25 September 2026 22:57

REAL TENDER TEXT - HELD-OUT TEST

252 real Oil India and NTPC lines, never looked at while building

0 wrong

of 4 merges made with no person involved

Item type read correctly **97.9%**

Two or more facts read **90%**

Duplicates found, with a reviewer **83%**
49 of 59 true pairs

Proposals that were right **18%**
the weak number: a terse line fits several items

Labels so far: one labeller — the builder, who had read these rows before this code was frozen. No fix was made from them. Two-person labels replace them. First freeze (24 September 2026): 0 wrong of 4, 86% found.

data/output/real_metrics.json · 25 September 2026 23:00

DEMO SET - 100 RECORDS

The records behind every page of this portal

0 wrong

of 141 merges made with no person involved

Duplicates found, with a reviewer **100%**

100 records — national codes **15 (truth 15)**

Unsafe merges refused **97**

We wrote this set's noise ourselves, so its 100% measures how well we undo our own errors — which is why the real test sits beside it.

data/output/metrics.json · 25 September 2026 23:00

Text matching on the same 15,000 records — each method at its best setting, tuned on one half and scored on the half it never saw

METHOD	MERGED	WRONG	TRAPS MERGED	RIGHT
Fuzzy string match	3,334	3,120	1,468	6%
TF-IDF	2,112	1,953	1,121	8%
Sentence embeddings	4,068	3,783	1,697	7%
Ours — specifications, automatic merges	217	0	0	100%

No text threshold reaches zero wrong: different items can share identical text once a 40-character field has cut them. Ours finds 18% of duplicates with no person involved — about what the text matchers find at their best, with none of their wrong merges — and sends the rest to a reviewer.

The synthetic sets measure how well the system undoes noise we generated. The real test is the more believable number, and it is reported beside them, never blended. Built 25 September 2026 from commit 2ed0138.

screen-evidence-panel.png — the portal's dashboard: three datasets side by side, never blended, each stamped with where its numbers came from. The weak number is shown with the strong ones.

INCOMING DESCRIPTION

a storekeeper is about to create a code for this

BEARING 6205 2RS 30X62X16MM

CONTRADICTS Contradicts itself

15 codes checked

One of the facts in this line is wrong, and nothing in it says which. No code is matched or issued until the line is corrected.

dimensions 30 x 62 x 16 mm — ISO 15: a 6205 is 25 x 52 x 15 mm - these dimensions are a 6206

screen-gate-contradicts.png — the creation gate stops a request that contradicts itself (6205 with a 6206's size) and refuses both codes it could be confused with.

EXTRACTED SPECIFICATIONS

ONGC	.	BPCL
* gasket type SPIRAL WOUND	✓	gasket type SPIRAL WOUND
* material grade SS316	✗	material grade SS304
VETO FIELD — INSTANT ZERO, WHATEVER ELSE AGREES		
* nominal size in 4	.	nominal size in 4
* pressure class 150#	.	pressure class 150#
* hard blocker — a mismatch here is an instant zero · not scored — the veto decided this pair		

WHAT IF

If the material grade matched, this pair would score **0.956** — an automatic merge.
Everything else agrees. One field decided — the rules refused it, not a threshold.

0.956

if equal · actual 0

screen-block-whatif.png — one field decided: "if the material grade matched, this pair would score 0.956".

PUT ON THE SLIDE (COMPACT LIST)

- **Problem context:** DPE Public Enterprises Survey 2023-24 (272 operating CPSEs) · SAP KBA 1630702 (MAKT-MAKTX, 40 characters) · GeM procurement Q1 FY25 · CPCL Manali refinery (10.5 MMTPA)
- **Standards used:** ISO 15 (rolling bearings) · ASME B36.10M (steel pipe) · ASME B16.5 (flanges) · IS 1367 (Part 3) = ISO 898-1 · IS 1364 (Part 1) = ISO 4014 · ASTM F3125 (replaces A325 / A490) · UNSPSC
- **Methods:** Fellegi & Sunter, *A Theory for Record Linkage* (1969) · Clopper & Pearson, *Biometrika* 26 (1934)
- **Industry evidence:** EY — Transforming MRO material master data (oil & gas case study) · CODA — material master cleansing case study
- **Data:** CPSE tender text from the Hugging Face dataset *Prasenjeet25/sih26099-cpse-material-codes*, licensed CC-BY-4.0
- **Images:** Rajiv Nair (EIL), CC BY-SA 4.0 · R. Henrik Nilsson, CC BY 4.0 · Bitjungle, CC BY-SA 4.0 — via Wikimedia Commons
- **Prototype:** our repository (add the GitHub link when the repo is public)

The Hugging Face credit is required by its CC-BY-4.0 licence, and the photo credits by theirs — keep all of them on this slide.

REFERENCES — FULL, WITH LINKS

FACT USED	SOURCE
272 operating CPSEs (448 reported)	Department of Public Enterprises, Public Enterprises Survey 2023-24 — dpe.gov.in
SAP material description MAKT-MAKTX is 40 characters	SAP KBA 1630702 — support.sap.com ; sapdatasheet.org
CPSEs: over ₹91,000 crore on GeM in Q1 FY2024-25	PSU Watch (PTI), 11 July 2024 — psuwatch.com
CPCL: IndianOil subsidiary since 2001; Manali 10.5 MMTPA	cpcl.co.in ; iocl.com
6205 bearing = 25 × 52 × 15 mm (ISO 15)	SKF 6205 — catalogue listing
4" SCH 40 pipe: OD 114.3 mm, wall 6.02 mm	ASME B36.10M — engineeringtoolbox.com ; Nominal Pipe Size
IS 1367 (Part 3) identical with ISO 898-1	BIS text (archive.org) ; ISO 898-1:2013
IS 1364 (Part 1) identical with ISO 4014	IS 1364-1:2002 (law.resource.org) ; ISO 4014
ASTM A325 / A490 withdrawn 2016, now grades of F3125	Portland Bolt — ASTM F3125; ASTM A325
UNSPSC: 8 digits, 4 levels; 31171500 = Bearings	U.S. Dept of Commerce; UNSPSC
SS316 has 2–3% molybdenum; better pitting resistance	ASTM A240; SS316 vs SS304 valve guide
Record linkage theory: link / possible link / non-link	Fellegi & Sunter (1969), JASA — overview
Exact binomial bound (0 wrong of 860 → < 0.35%)	Clopper & Pearson (1934), <i>Biometrika</i> 26:404 — OUP
~10% potential MRO savings; duplicates in oil & gas MRO data	EY case study — ey.com
~5–7% inventory cost reduction after standardisation	CODA Technology Solutions case study — codasol.com (vendor case — always attribute)
Real CPSE tender text, CC-BY-4.0	Hugging Face — Prasenjeet25/sih26099-cpse-material-codes
SIH evaluation criteria	SIH 2026 guide ; official template's instructions slide

IMAGE LIBRARY

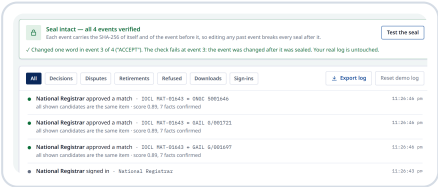
all files are in the images folder next to this PDF

FILE	USE ON	WHAT IT IS	CREDIT / LICENCE
architecture.png / .svg	Slide 3	The full pipeline, three lanes	Ours
inversion.png	Slide 2	Why text matching fails — the core idea	Ours
numbers.png	Slide 4	Headline numbers strip	Ours
text-vs-specs.png	Slide 4	Text matchers vs ours on the same pairs	Ours
screen-review-queue.png	Slide 3	Prototype: review queue	Ours
screen-real-text.png	Slide 3 / 2	Prototype: real tender text with evidence	Ours
screen-evidence-panel.png	Slide 5 / 4	Prototype: dashboard evidence panel	Ours
screen-gate-contradicts.png	Slide 5	Prototype: creation gate	Ours
screen-block-whatif.png	Slide 2 / 5	Prototype: refused merge with its what-if	Ours
screen-audit-seal.png	Slide 5 (optional)	Prototype: sealed audit log and tamper test	Ours
photo-refinery.jpg	Slide 1 (optional)	Vizag refinery modernisation, reactor erection	Rajiv Nair (EIL), CC BY-SA 4.0, Wikimedia Commons
photo-ball-bearing.jpg	Slide 2 (small)	Deep groove ball bearing	R. Henrik Nilsson, CC BY 4.0, Wikimedia Commons
photo-gate-valves.jpg	Slide 2 (small)	Three flanged gate valves	Bitjungle, CC BY-SA 4.0, Wikimedia Commons

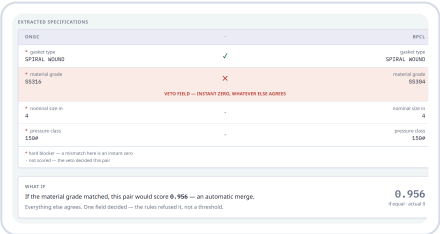
Not included on purpose: CPCL, IOCL, MoPNG or Government of India logos and the State Emblem — they are protected marks and our prototype must not look official. Use only the SIH template's own logos.



photo-refinery.jpg



screen-audit-seal.png



screen-block-whatif.png

QUESTION	ANSWER
Why not just buy SAP MDG or Informatica?	They are built for one enterprise and need all data in one place. No CPSE will hand over its prices and vendor list. Ours works federated — only bands and hashes cross — and it is an Indian registry for national procurement data.
Isn't this just string matching?	The opposite. Text matching scores SS316 vs SS304 at 0.989 and would merge them. We match on specifications and veto on grade — and on the same 15,000 records the text methods are 6–8% right while ours made 0 wrong merges.
Your data is synthetic — why believe it?	Fair — it measures how well we undo noise we generated. So we test on 252 real NTPC and Oil India lines held back while building: 0 wrong merges, item type 98%. We report both, never blended.
What if the AI hallucinates a grade?	The LLM only reads — it never decides a merge. The rule reader runs first and every value it takes keeps the words it came from; a record that contradicts itself is held for a person; the rules decide. Next phase: the LLM must quote its words, and a value whose quote is not in the line is dropped.
Who approves a merge between two companies?	The national registrar links codes; any affected company can dispute at any time and the link returns to review; only the owning company can retire its own code. Every step is in a sealed audit log.
How does it scale to millions?	One LLM call per record, stored; pairwise scoring is arithmetic. Blocking skips 99.8% of pairs. At national scale: PostgreSQL + pgvector.
Is 150# the same as 150 psi?	No — 150# is an ASME B16.5 pressure class; a Class 150 carbon-steel flange is rated around 285 psig at ambient and less when hot. That is why pressure class is a veto field.
What is your weakest number?	Review precision: about one in five pairs sent to a reviewer is a true duplicate, because a short line fits several items. So the reviewer sees one record with its three best candidates — the right one is there about four times in five.
Will CPSEs actually change their codes?	They don't have to. The national code sits above their codes and links them; every original code stays in use. Phase 1 cleans duplicates inside one company with no data sharing at all.

DO NOT SAY

- “100% accurate” or “100% precision” — say **“0 wrong out of 860”** and **“below 0.35% at 95% confidence”**.
- “Production-ready”, “live SAP integration”, “secure”, “encrypted” — say **prototype, SAP-shaped export file, reduces data exposure**.
- “Material master” about the real data — it is **tender text from two CPSEs**.
- “Descriptions are cryptic because SAP caps them at 40 characters” — give the several-reasons version on slide 2.
- Any synthetic number without the real one beside it; any saving as a promise rather than “published case studies report”.
- That named vendors supply CPCL — that is not public information.

IF THERE IS A DEMO — 90 SECONDS

SECONDS	SHOW	SAY
0–15	Real Tender Text page: 101.60 mm vs 4"	"This is a real government tender. Only the weight proves these are one pipe."
15–30	Case: The Match — SKF vs FAG 6205	"Few shared words. The standard says they are the same bearing."
30–45	Case: The Block — SS316 vs SS304	"Near-identical text. Grade is a veto."
45–60	The what-if line	"If the grade matched, this would score 0.956. One field decided."
60–75	Creation gate — the contradicting 6205	"It contradicts itself. Caught before it can match anything."
75–90	Audit trail — Test the seal	"Change one word and the seal breaks. Every decision can be replayed."

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